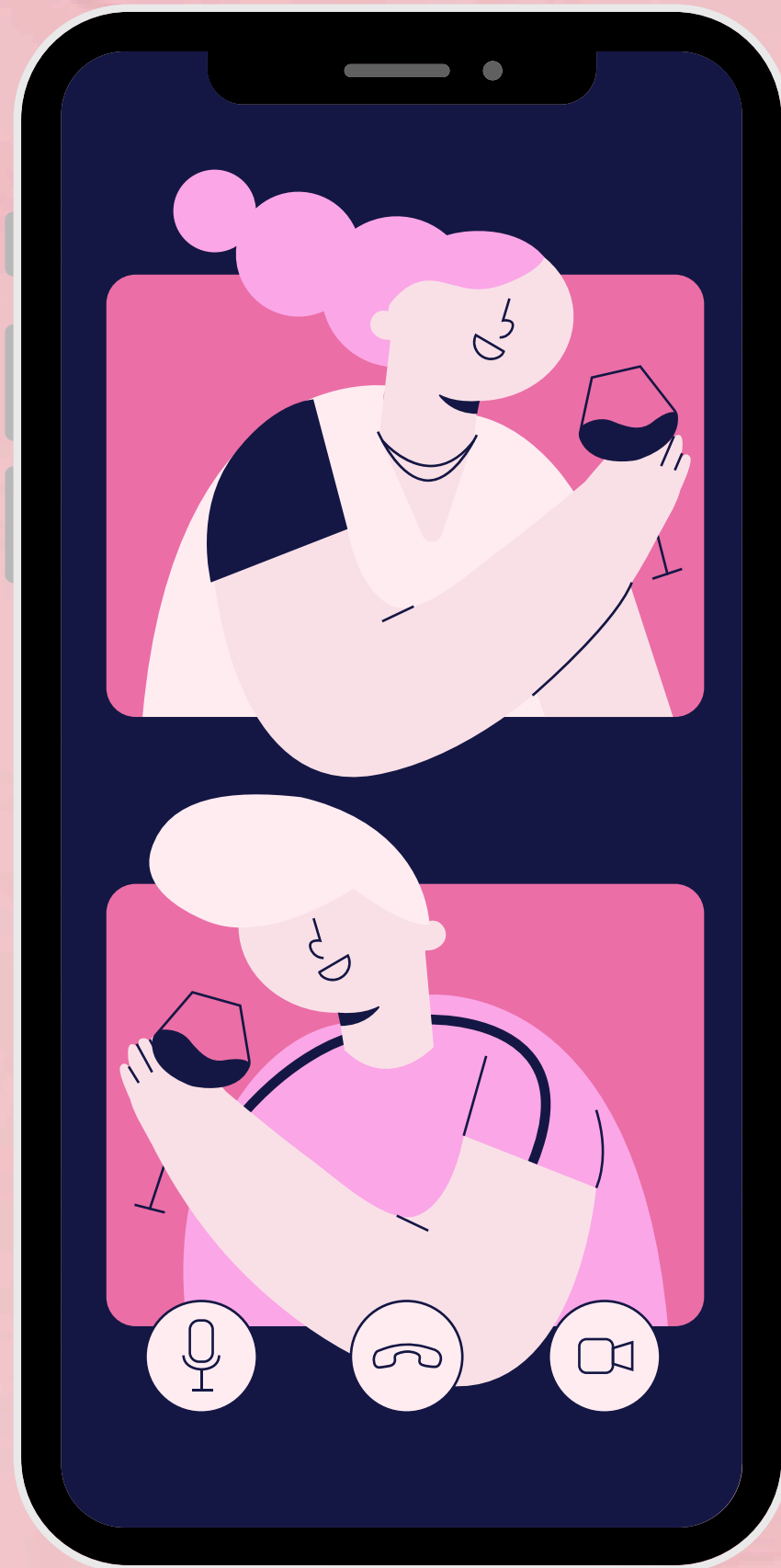


MGTA457 :
Business Intelligence Systems

Beyond the Swipe

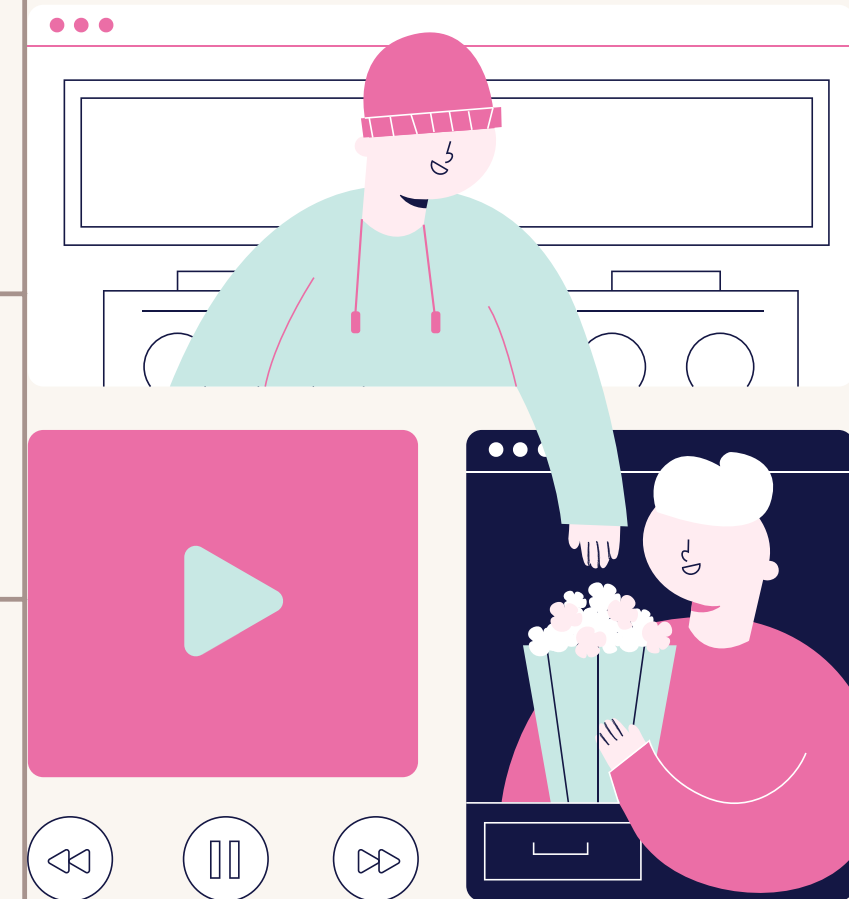
~ an insight into dating apps

Prabhlín Kaur Matta, Abhigna Akkipeddi, Chandana Vuppu,
Namratha K, Pavithra Vaidyanathan



Agenda

01	Project Goals
02	Data
03	Dashboard
04	Key Takeaways
05	Conclusion



Project Goals

Motivation and Objective

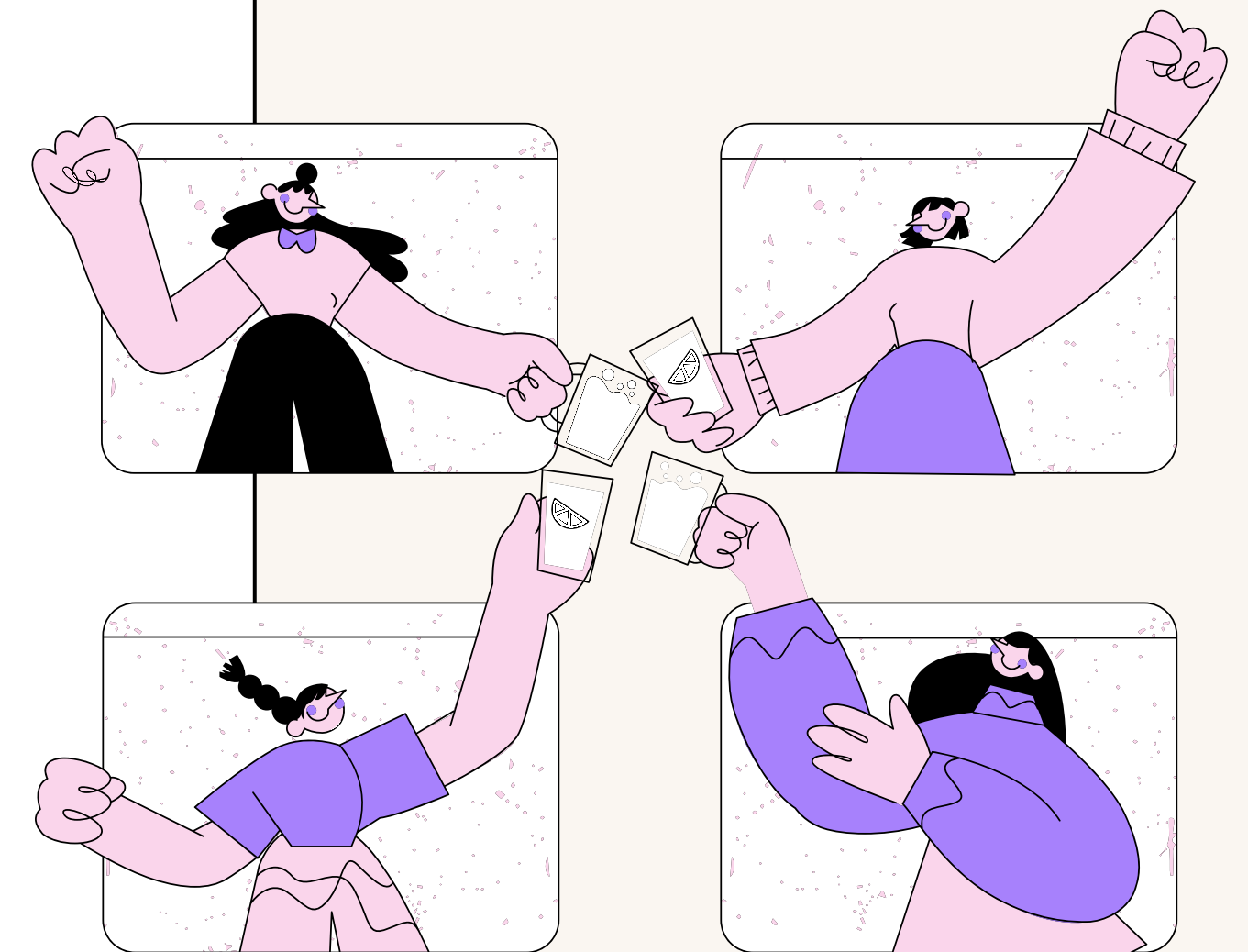
- Dating apps contain rich behavioral data that reveal how users present themselves and interact.
- Understanding user personas helps improve matching, personalization, and overall user experience.
- Our goal is to build dashboards that turn raw data into a clear story of who users are, how they behave, and what drives engagement.



Project Goals

Targeted Audience

- Product teams seeking to improve user experience and matching quality
- Marketing & UX teams needing clear user personas
- Users wanting insights to optimize their profile and engagement



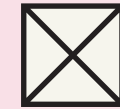
Data



Data Source

kaggle

*csv files containing user profiles,
behaviors, and engagement metrics



Data Prep

Tableau Prep



Used user ID, and joined profiles with
behavior into one analysis-ready
dataset.

Dashboard

~ Beyond the Swipe



Average Likes
Recieved

99.53

Average
Messages Sent

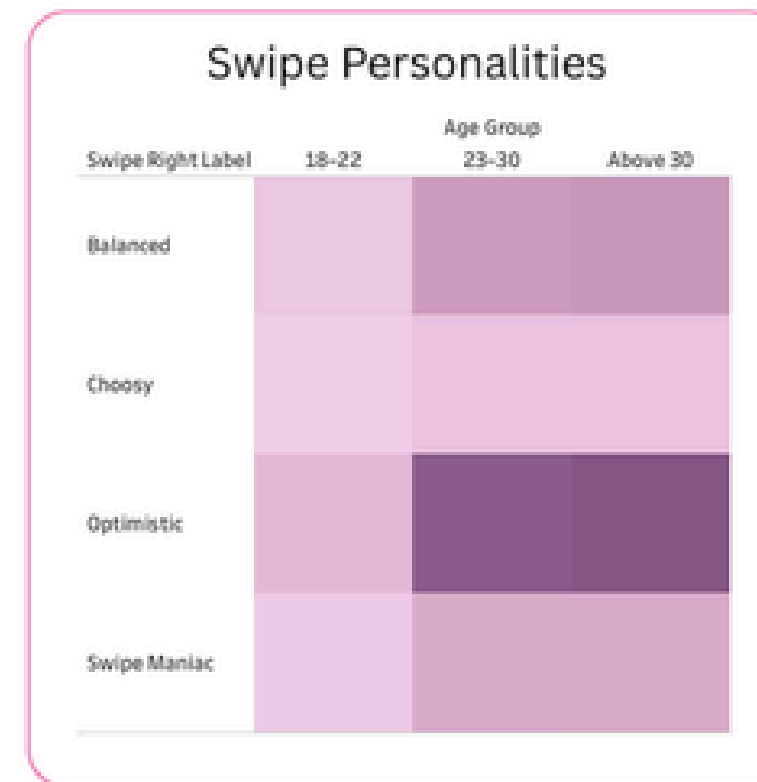
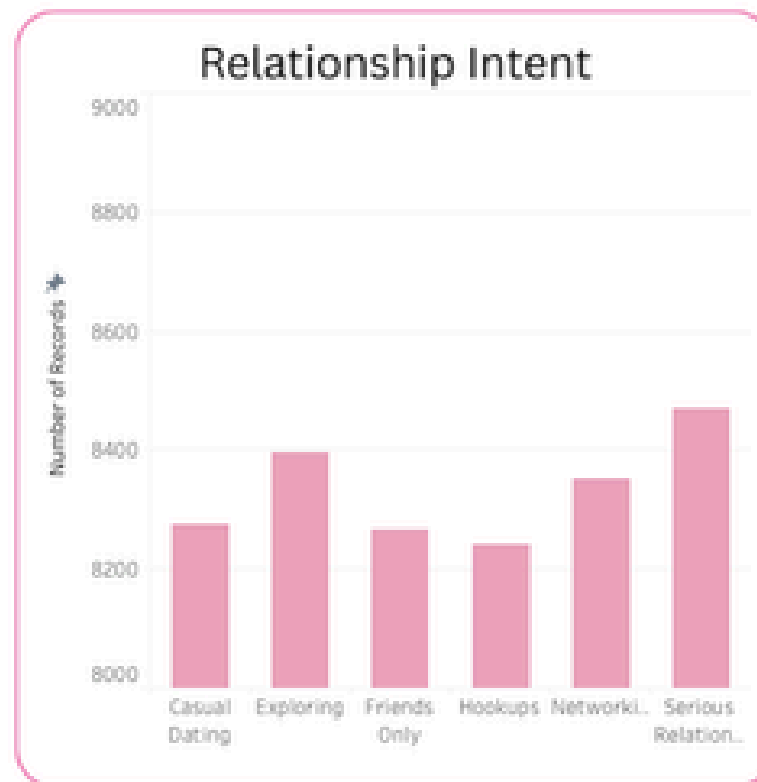
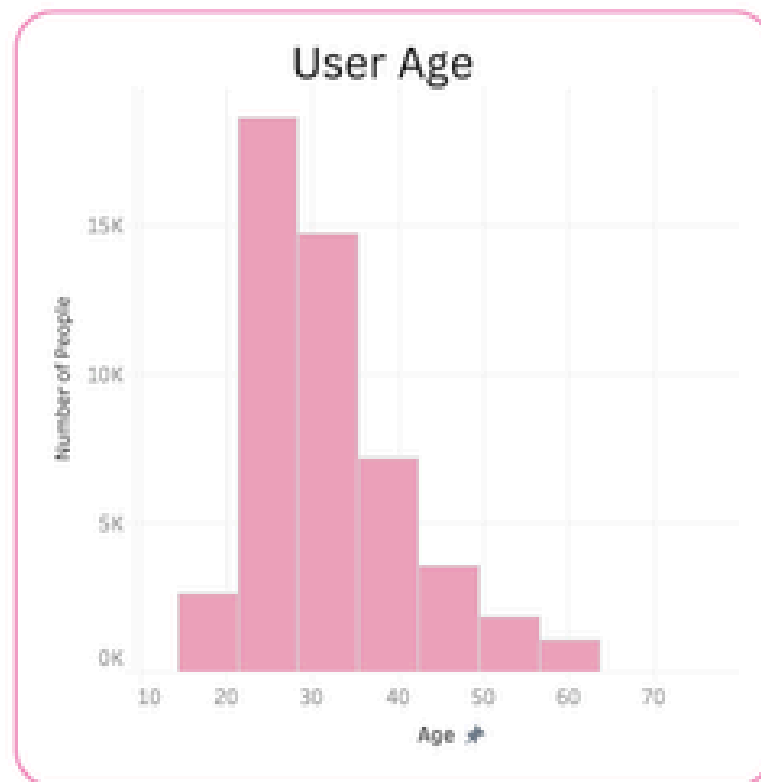
50.07

Average
Time Spent

149.9
mins

Most Active
Time of Day

After Midnight



Key Takeaways

Late-night browsing is when users are most engaged.

Most users fall between ages 18—30, with a balanced sex split.

Relationship intentions range from casual to serious, i.e a diverse user base with varied expectations.

Swiping behavior shifts with age: younger users swipe more selectively, while older groups show more optimism, balanced patterns—highlighting different decision styles across age ranges.

Profile Optimization Insights: How Profile Choices Influence Likes

Data-driven analysis of how photos, bios, and verification impact user engagement.

Verified users get
2.2×
as many likes

3-5 Photos
> 200K Likes
Less is More!

3× More Likes
for complete profiles
Detail is key

Short bios get
2× likes
Keep it short and sweet!

Likes vs Pics

Analyzes how the number of uploaded photos affects total likes received



Bio Strength

Measures the impact of having a bio on engagement



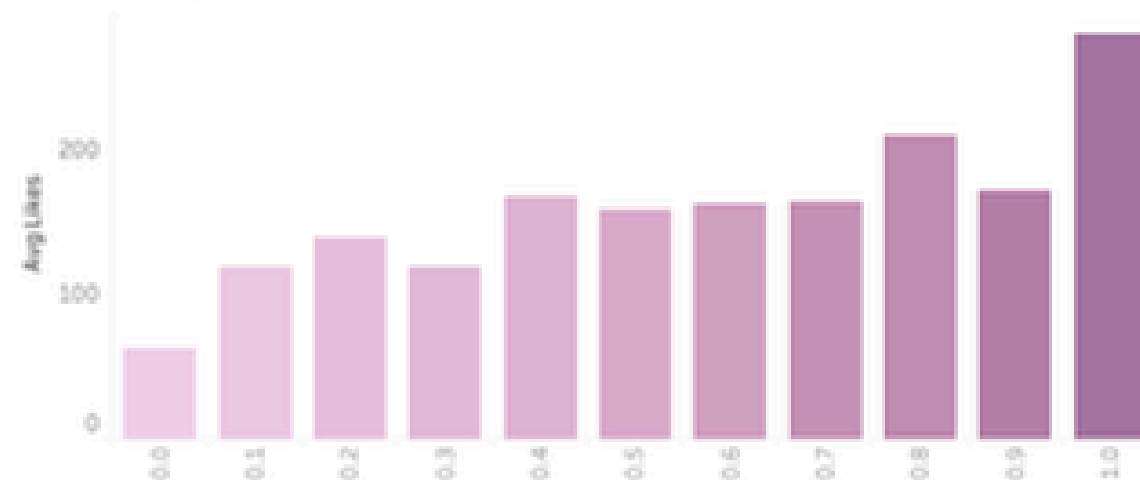
Bio Length vs Likes

Evaluates whether short, medium, or long bios generate more likes



Profile Strength

Shows how overall profile completeness correlates with average likes.



Verified Users vs Likes

Comparison of engagement levels between verified and non-verified users.



- Users with 3–5 photos get the most likes, outperforming both low-photo and high-photo profiles.
- Having any bio triples likes, but short bios deliver the best performance, beating medium and long ones.
- Verified users receive 2.2× more likes than non-verified users, highlighting the importance of trust signals on the platform.

When Users Swipe: Activity & Session Timing Insight

Analyzing hourly rhythms and session behaviors to reveal when users are most active.

12 PM

peak activity,
peak hope

**Afternoon
& Midnight**

Peak Swipe Hours

**10-15 min
Sessions**

Short sessions,
strong connections...

**14:00
-22:00**

Prime time for prime matches

24-Hour Behavioral Rhythm of User Engagement

Radial clock signifying the user's last active hour



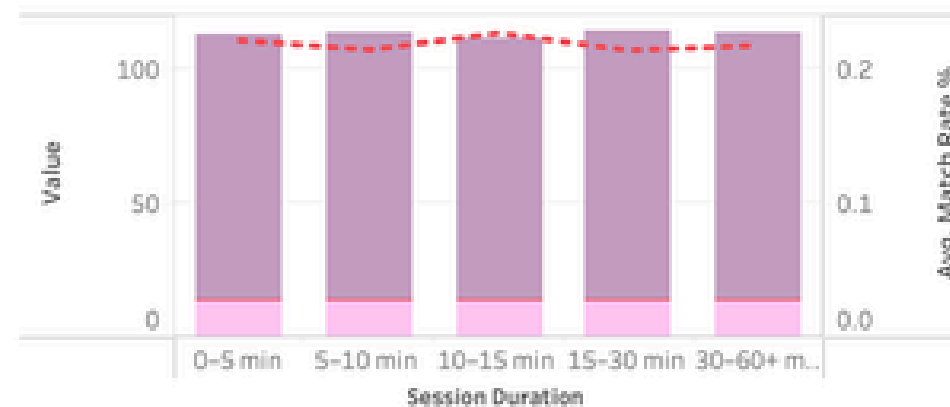
Measure Names

Avg. Likes Received
Avg. Match Rate %
Avg. Matches (Cleansed)

Count of Hour of Day (0-23)
2,012 2,137

Impact of Session Duration on User Engagement

Match Success Rate during the day



Swipe Intensity Across Time of Day

Heatmap showing swipe intensity across 24 hrs

	Hour of Day (0-23)							
Swipe Ti..	0	1	2	3	4	5	6	7
After Mi..	342	327	325	353	334	395	363	350
Afterno..	345	311	351	372	377	358	308	355
Early M..	352	349	360	361	364	332	366	307
Evening	334	348	324	350	355	363	355	324
Late Nig..	345	365	357	329	338	328	370	341
Morning	328	312	350	372	350	349	347	351

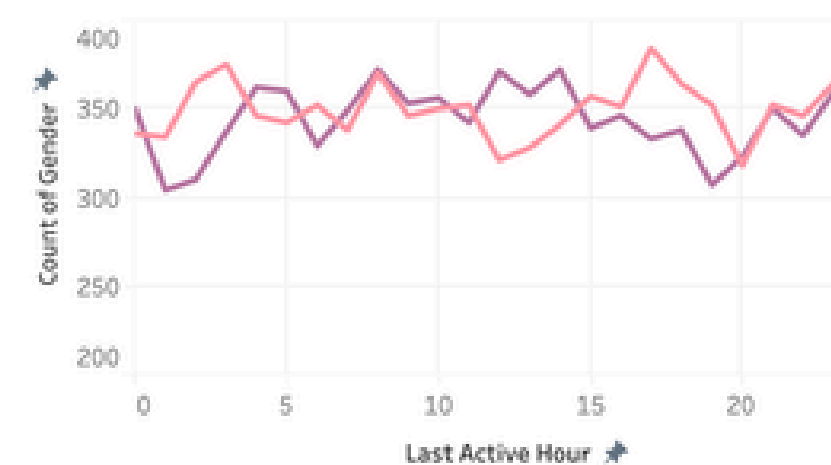
Gender

Female
Male

Count of Number of Records
303 421

Hourly Activity Patterns By Gender

Who's Online When?



- User activity peaks at 12 PM, making midday the best window for visibility.
- Swipe activity is highest late at night, when users swipe more heavily.
- Sessions of 10–15 minutes produce the highest match rates.
- Female engagement is steadier, while male engagement shows sharper peaks.

Dating App Churn Story: From Swipe to Relationship

A behavioral view of how users move through the dating funnel, where they drop off, and which engagement signals are linked to retention.

Overall Churn Rate

60%

High Engagement

27%

Avg Mutual Matches Per User

13.87

Simulated Risk Tier

High Risk

Low Risk

Medium Risk

Churn Status

(All)

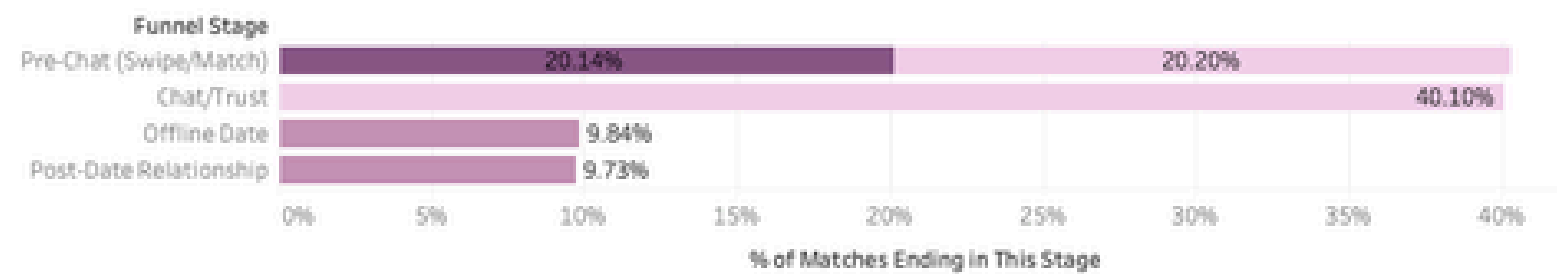
Churn

Engaged

Neutral

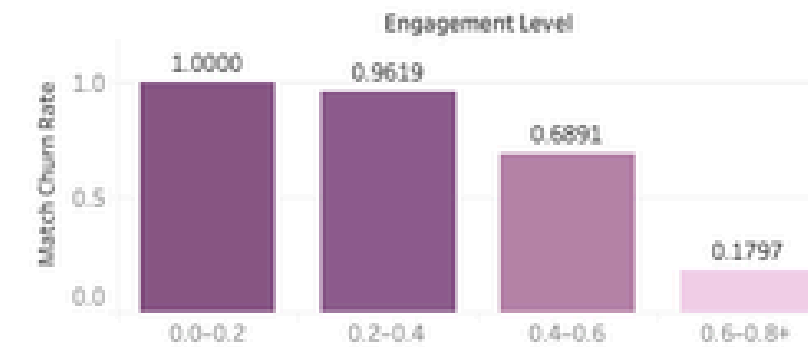
Funnel by Stage – Where Outcomes Happen

Most matches stall in the early stages; few progress to offline dates or relationships.



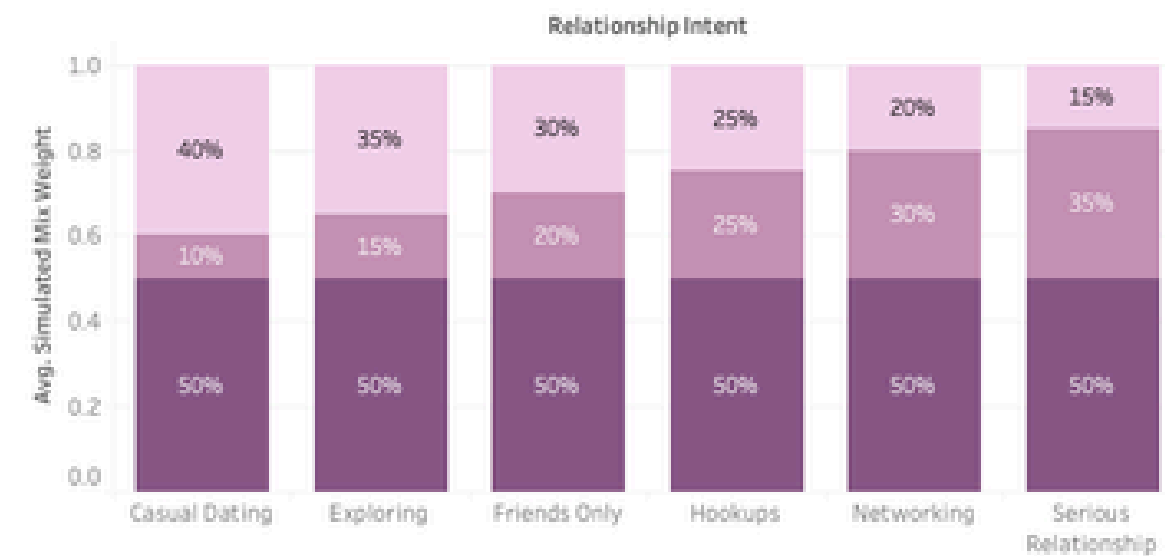
Churn vs Engagement Level

Churn is highest for low-engagement users and lowest for highly engaged users.



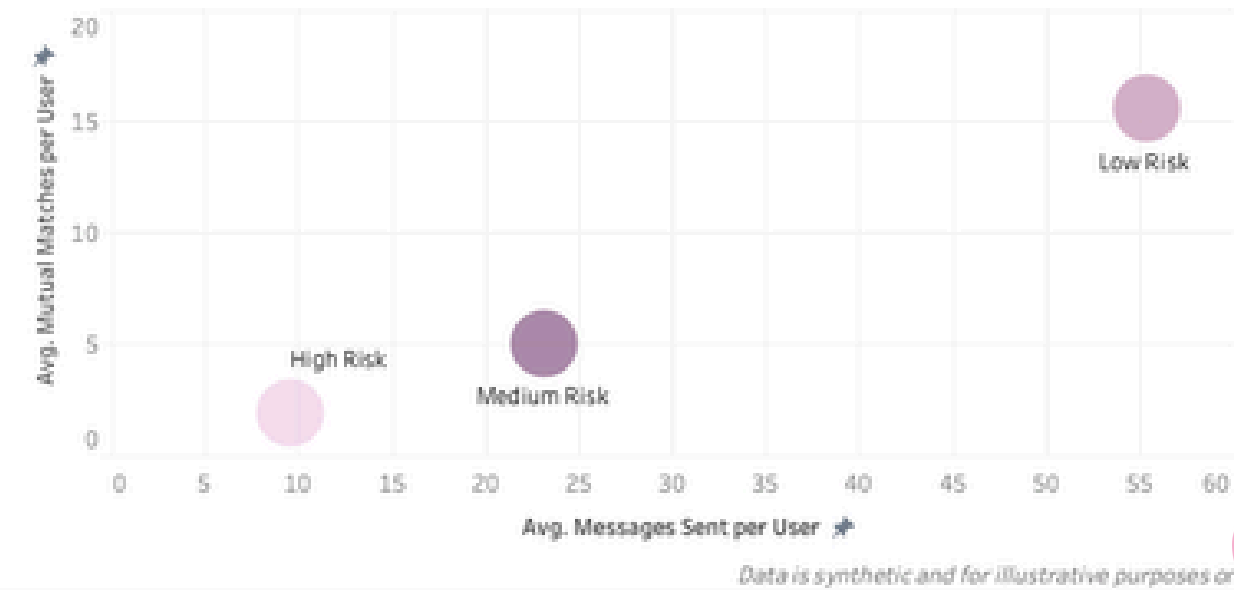
Churn Mix by Relationship Intent

Within each intent, how many users churn, stay neutral, or move toward engagement?



Messages vs Matches by Risk Tier

Are low-risk users messaging more and getting more matches than high-risk users?



- Biggest leak is early — Most matches stall in swipe + early chat; very few reach dates or relationships.

- Engagement drives success — Users who stay active (more engagement + messaging) churn less and earn more mutual matches.

- PM and UX can use this dashboard to spot where matches are dying and which behaviors define healthy users, then design experiments to push more people into that “healthy” pattern.

4

♡ THE
END ♡

